Binocular Rivalry

Cecilia Galbiati, Giulia Di Vincenzo, Ferran Estrella, Gerard Carol, Arnau Ruiz

December, 2025

1 Introduction

Binocular rivalry is a phenomenon of perceptual bistability, where presenting different images to each eye does not result in a merged perception. Instead, perception alternates between the two images over time. This perceptual alternation has been a focal point in cognitive neuroscience, providing insights into how the brain processes conflicting sensory information.

The underlying mechanisms of binocular rivalry shed light on neural competition, adaptation, and the dynamics of visual processing. These alternations are influenced by various factors, including the strength of the visual stimuli, inherent noise, and inhibitory interactions between neural populations. Studying these dynamics is critical for developing theoretical models of perceptual decision-making and understanding broader brain functions related to conflict resolution and sensory integration.

In this report, we investigate a computational model which describes the oscillatory dynamics and stability of neural populations involved in binocular rivalry. Through exploration of parameter variations, such as stimulus strength and noise, we seek to study the model's behaviour compared to real-life behaviour, particularly focusing on the rate of perceptual alternations and dominance duration distributions.

2 General Binocular Rivalry Model

Let us consider two neuronal populations whose activity corresponds to the presentation of mutually incongruent images to the left and to the right eye. The neuronal populations are modelled to compete through mutual inhibition, while a slow fatiguing mechanism, referred to as adaptation, mediates the back-and-forth switching of dominance.

Precisely, the continuous dynamical system describing the firing rates of these neuronal populations, which best captures the oscillatory behaviour in binocular rivalry, is of the form:

$$\tau \frac{dr_1}{dt} = -r_1 + \sigma(-\gamma a_1 - \beta r_2 + I_1 + \eta_1), \tag{1}$$

$$\tau \frac{dr_2}{dt} = -r_2 + \sigma(-\gamma a_2 - \beta r_1 + I_2 + \eta_2),$$

$$\tau_a \frac{da_1}{dt} = -a_1 + r_1, \tag{2}$$

$$\tau_a \frac{da_2}{dt} = -a_2 + r_2,$$

where, for each population $j \in \{1, 2\}$: $r_j(t)$ denotes the firing rate and τ is the time scale of the firing-rate dynamics; $a_j(t)$ represents the adaptation variable with time scale τ_a; γ and β are parameters representing the strengths of adaptation and inhibition, respectively; I_j is the input stimulus, and η_j is the noise term.

Additionally, σ denotes the transfer function given by

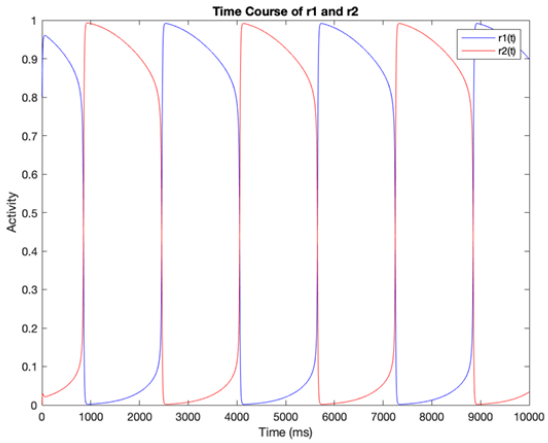
$$\sigma(u) = \frac{1}{1 + \exp\left(-\frac{u-\theta}{k}\right)}, \tag{3}$$

for some parameter values θ and k.

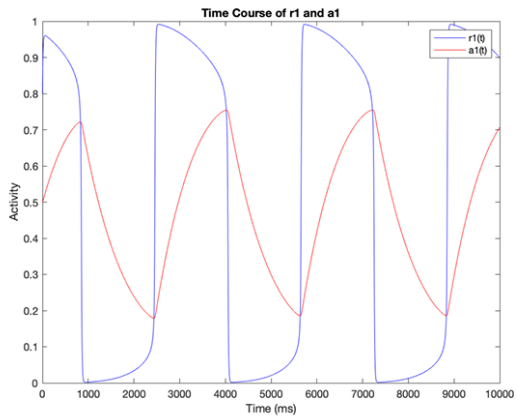
Throughout this document, we will take the parameter values $\tau = 10$ ms, $\tau_a = 1$ s, $\gamma = 0.5$, $\beta = 0.9$, $k = 0.1$, and $\theta = 0.2$. We will also consider a deterministic input stimulus taking the same value $I = I_1 = I_2$ for both neuronal populations. Additionally, from now until the last chapter, we assume that the model is deterministic and involves no stochastic noise, i.e., $\eta_j = 0$.

3 Phase Plane Analysis and Nullclines

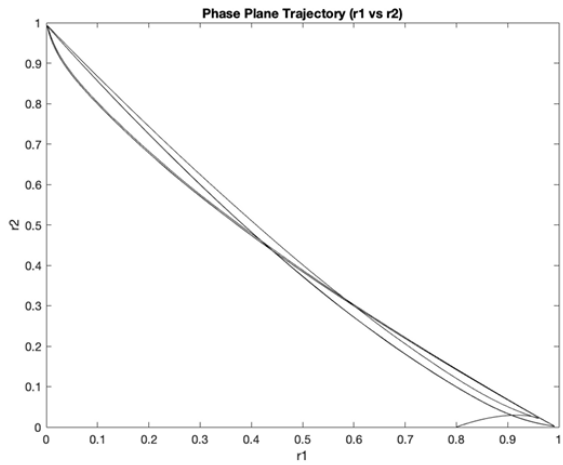
If we integrate the previous system of equations for a certain stimulus $I = 0.8$, we obtain the dynamics represented in Figure 1. We note that as the time courses for r_1 and r_2, we can observe a clear oscillatory behaviour. In the plot of the time courses of r_1 and a_1, we can see that as the adaptation variable increases, the firing rate rapidly decreases, as expected. The oscillatory dynamics can also be inferred from the phase plane trajectory, where we observe an oscillation between $r_1 = 1$ and $r_2 = 1$.



(a) Time course of $r_1(t)$ and $r_2(t)$



(b) Time course of $r_1(t)$ and $a_1(t)$



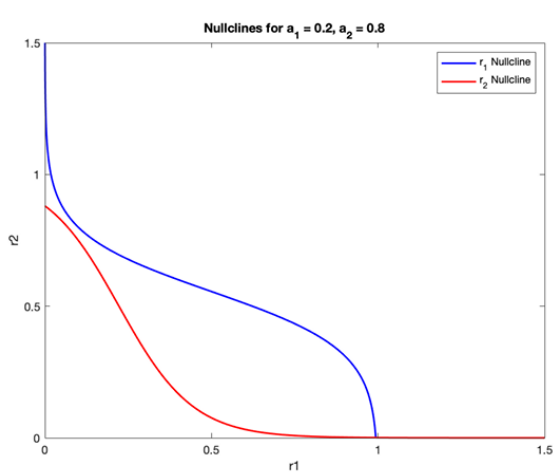
(c) Phase plane trajectory ($r_1(t)$ vs $r_2(t)$)

Figure 1: Representations of the neural dynamics in the model.

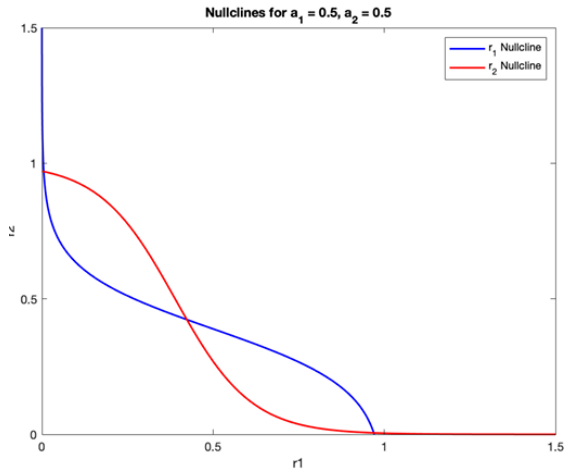
If we freeze the values of the adaptation variables a_j ("slow variables"), the system no longer exhibits oscillatory behaviour. To examine the stationary states under this assumption, we plot the nullclines of the system in Figure 2 for different fixed values of the adaptation variables.

We observe that when the adaptation variables are fixed at different values, the only equilibrium point of the system occurs when the r_j associated with the higher adaptation a_j is zero, while the other firing rate equals one. However, when both adaptation values are frozen at 0.5, there exists an equilibrium point in which both firing rates are non-zero. This would correspond to perceiving a merged image when each eye is shown a different image. This scenario, with a_j frozen at 0.5, is, however, not biologically plausible.

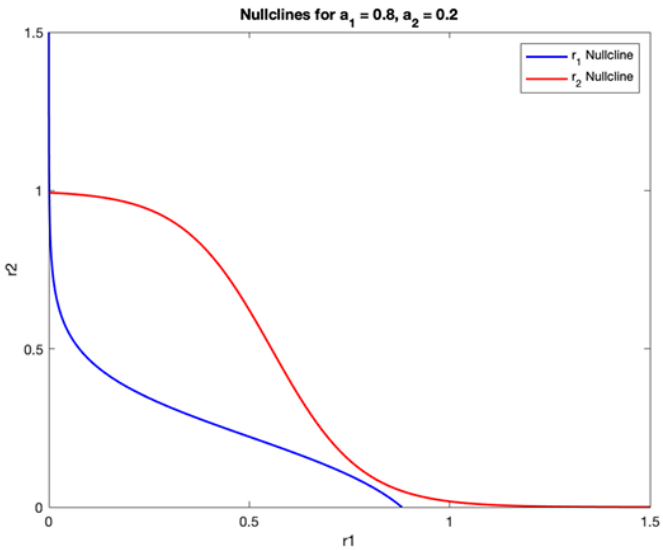
The displayed dynamics when the slow variables are frozen indicate that bistability appears only in the "fast dynamics" regime.



(a) Nullclines for $a_1 = 0.2$, $a_2 = 0.8$



(b) Nullclines for $a_1 = 0.5$, $a_2 = 0.5$



(c) Nullclines for $a_1 = 0.8$, $a_2 = 0.2$

Figure 2: Nullcline visualizations for different frozen values of the adaptation variables a_1 and a_2.

4 Oscillation Range and Period Dependence on Stimulus Strength

In this section, we show that the model oscillates for stimulus values in the range $0 \leq I \leq 2$, and we also analyse whether a stronger stimulus I decreases the oscillation period. Precisely, we examine the validity of the proposition by Willem Johannes Maria Levelt, which states that increasing the stimulus strength in both eyes, while keeping it equal between the eyes, will increase the perceptual alternation rate, i.e., decrease the oscillation period.

In Figure 3 we can see the oscillation period as a function of I according to our model. We observe that the oscillation period increases until $I \approx 0.9$ and decreases after that. Hence, Levelt's proposition holds for our model only if I is large enough.

To study why the period increases with I for low values and decreases for high values, we plot the minimum and maximum firing rates of one of the neuronal populations as a function of I, as shown in Figure 3.

We observe that for small I, the minimum and maximum firing rates are both low and close to each other. As I increases, the maximum firing rate rises, implying that the amplitude of the oscillations increases, and consequently, the period increases. The period reaches its maximum when the amplitude reaches its maximum. Beyond this point, as I continues to increase, the minimum firing rate also rises, causing the amplitude of the oscillations to decrease, and consequently, the period decreases.

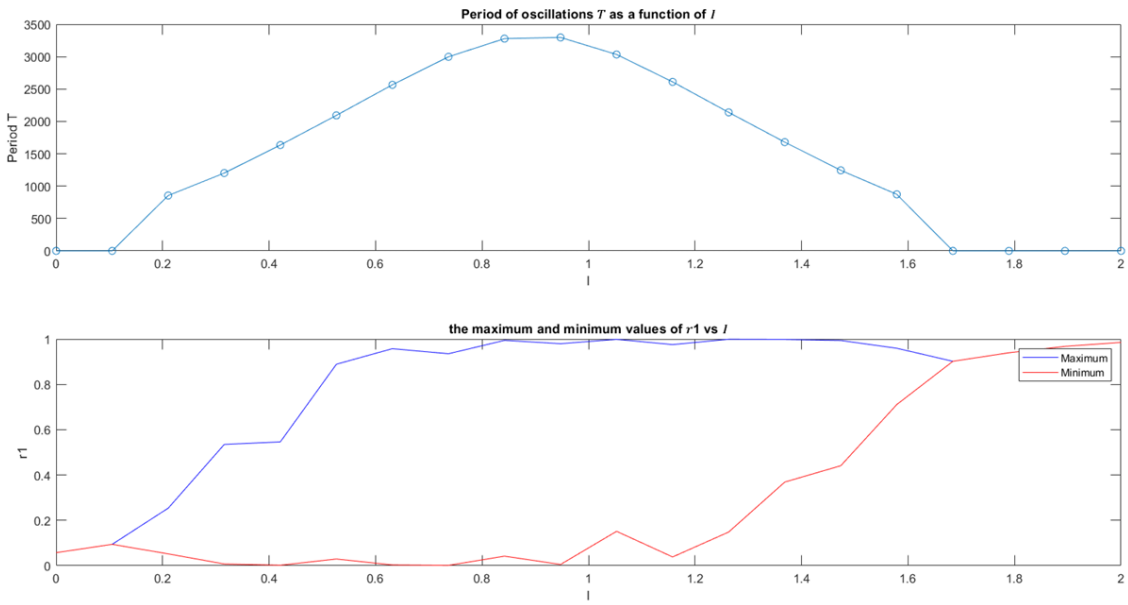


Figure 3: Above, the period of oscillations as a function of the stimulus strength I. Below, the maximum and minimum values of the firing rate r_1 as a function of I.

5 Oscillatory Behaviour Under Enhanced Mutual Inhibition

In this section, we increase the inhibitory parameter β from 0.9 to 1.1 to explore the effect of enhanced mutual inhibition between the two competing neural populations on the system's oscillatory dynamics. The model remains deterministic and is analysed for $0 \leq I \leq 2.5$. Specifically, we examine how the oscillatory period T and the amplitude A of the neural activity vary with increasing stimulus strength I for the increased β. We also discuss the possible Winner-Take-All (WTA) behaviour of the system for certain values of I.

In Figure 4, we plot the period T and the amplitude A of the system's oscillations as functions of I. We observe that Levelt's proposition holds approximately in the regions $0.5 \leq I \leq 0.75$ and $1.5 \leq I \leq 2$. Between the high peaks in the figures, in the intermediate range $0.75 \leq I \leq 1.2$, the period and amplitude of the oscillations drop close to zero. This indicates non-oscillatory behaviour, where one population dominates the other, referred to as a Winner-Take-All (WTA) regime.

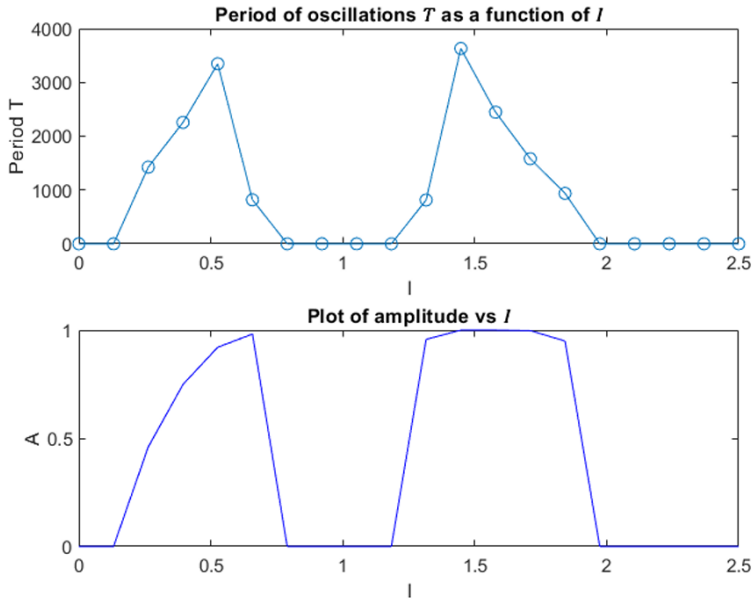


Figure 4: Period and amplitude of oscillations as functions of I.

In Figure 5, we show a phase-plane plot of the firing rates for $I = 1.0$, within the WTA regime. We observe that for the considered set of initial conditions, the stationary state corresponds to one in which population 2 dominates, with $r_2 = 1$ and $r_1 = 0$, and no oscillatory behaviour is observed. We remark that the plot displays trajectories in which the "winner" is always the same, but this is not true for all initial conditions. Instead, for values of I in the WTA regime, we observe bistability between dominance of populations 1 and 2, and depending on the initial conditions, the stationary "winner" can be either population 1 or population 2.

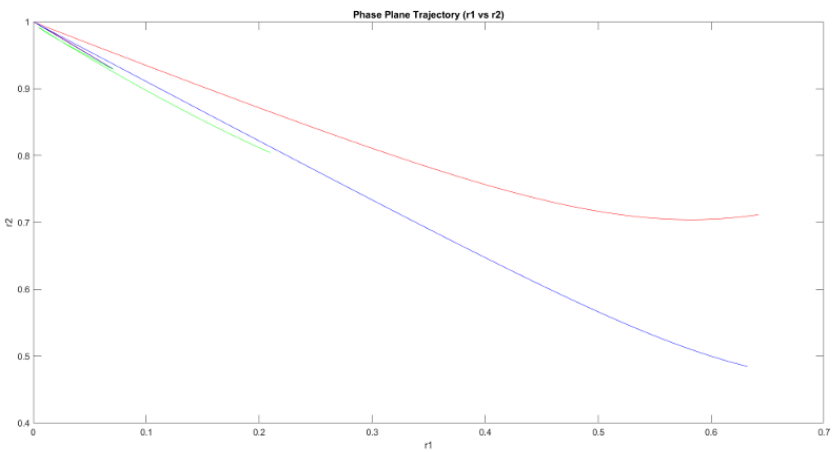


Figure 5: Phase plane trajectory when $I = 1.0$, using 3 different initial conditions.

6 Effect of Decreasing τ_a

In this section, we explain why decreasing τ_a causes the oscillations of the system to disappear.

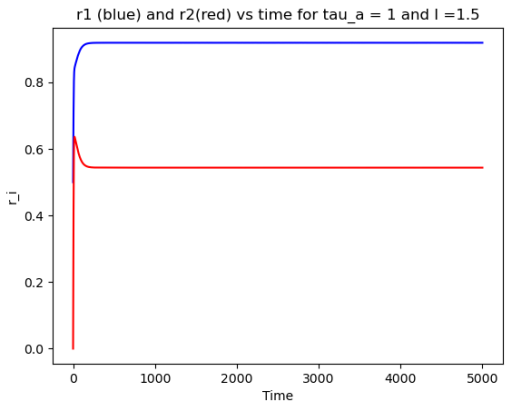
First, we recall that a_j are the slow adaptation variables, with associated time scale $\tau_a = 100\tau = 1s$. This means that the neuronal populations compete through mutual inhibition with a delay of $\tau_a = 1s$. Hence, decreasing the parameter τ_a is making the inhibition faster.

As $\tau_a \to 0$, directly from the dynamics (2), we have that $a_1 \to r_1$ and $a_2 \to r_2$. Thus, system (1) becomes:

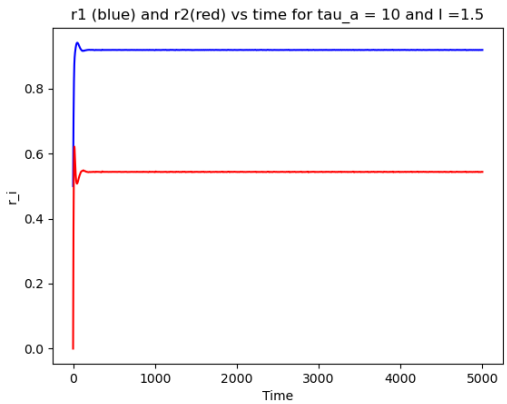
$$\tau \frac{dr_1}{dt} = -r_1 + \sigma(-\gamma r_1 - \beta r_2 + I_1), \tag{4}$$
$$\tau \frac{dr_2}{dt} = -r_2 + \sigma(-\gamma r_2 - \beta r_1 + I_2).$$

The system now becomes an I–I network, following the equations of a mutually inhibitory system with instantaneous self-inhibition γ and cross-inhibition β. In an I–I network, no oscillations occur, since it operates in a WTA regime where r_1 and r_2 compete directly.

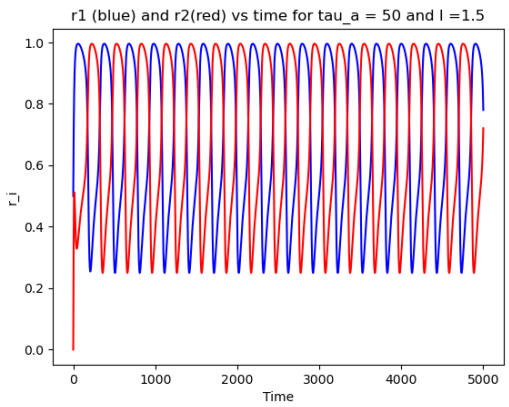
To conclude this section, we illustrate in Figure 6 how decreasing the τ_a parameter causes the system to transition from an oscillation regime to a WTA regime.



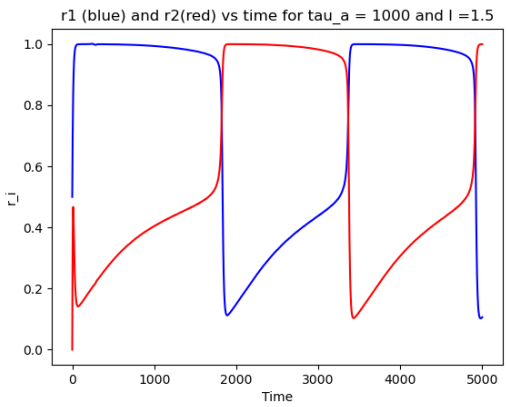
(a) WTA regime for $\tau_a = \tau$



(b) WTA regime for $\tau_a = 10\tau$



(c) Fast oscillation regime for $\tau_a = 50\tau$



(d) Typical oscillation regime for $\tau_a = 1000\tau$

Figure 6: Visualizations of the oscillations of r_1 and r_2 with respect to different values of the parameter τ_a, for $I = 1.5$. We observe that when τ_a is on a similar time scale as τ, the firing rates do not oscillate. However, when τ_a is increased substantially relative to τ, oscillations appear.

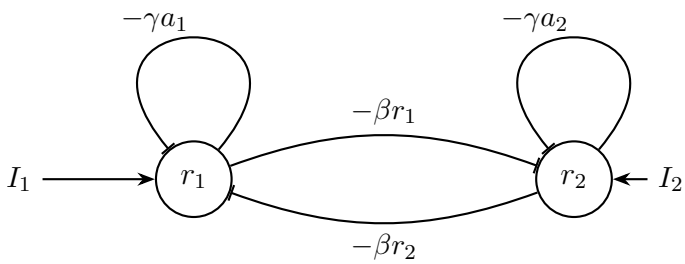


Figure 7: Schematic of a general binocular rivalry network. Each population activity, r_1 and r_2, receives an external input I_1 and I_2, and they mutually inhibit each other with strength β. Each unit also exhibits self-inhibition through an adaptation mechanism a_i, weighted by γ. Slow adaptation mediates rhythmic alternations between dominant states (oscillations), whereas fast adaptation leads to a WTA regime (bistability).

7 Equilibrium Point in the Symmetric Case

What would happen if we had two perfectly symmetric firing rates with the same slow adaptation? Would this lead to an equilibrium point, and how would its stability behave? In this section, we address all of these questions.

Let us assume that $r_1 = r_2 = r$, $a_1 = a_2 = a$ and also $I_1 = I_2 = I$. In this case, the system of equations (1) and (2) reduces to:

$$\tau \frac{dr}{dt} = -r + \sigma(-\gamma a - \beta r + I), \tag{5}$$

$$\tau_a \frac{da}{dt} = -a + r. \tag{6}$$

The necessary and sufficient conditions to have an equilibrium point of the system are $\frac{da}{dt} = 0$ and $\frac{dr}{dt} = 0$. Applying the condition $\frac{da}{dt} = 0$ to (6) yields the necessary condition $a = r$. Then, applying $\frac{dr}{dt} = 0$ to (5) leads to the equation:

$$\sigma(I - (\gamma + \beta)r) - r = 0. \tag{7}$$

If we denote by $c = \exp(-\frac{I-\theta}{k}) > 0$ and $d = \frac{\gamma+\beta}{k} > 0$, then finding the equilibrium points of the system in the symmetric case $(r_1, r_2, a_1, a_2) = (r, r, a, a)$ reduces to imposing $a = r$ and finding a zero r^* of the function

$$h(r) = \frac{1}{1 + c\exp(dr)} - r. \tag{8}$$

Now, we note that:

- We have $h(0) = \frac{1}{1+c} > 0$ and $h(1) = \frac{1}{1+c\exp d} - 1 < 0$, which implies by Bolzano's theorem that $\exists r^* \in (0, 1)$ such that $h(r^*) = 0$.
- Since $h'(r) = -\frac{cd\exp(dr)}{(1+c\exp(dr))^2} - 1 < 0$ for all r, then h is monotonically decreasing and therefore r^* is the unique zero of h.

Hence, there exists a unique equilibrium point $(r_1, r_2, a_1, a_2) = (r^*, r^*, r^*, r^*)$ of the system in the considered symmetric case.

7.1 Characteristic Exponents of the Equilibrium Point

To study the linear stability of the equilibrium point $(r_1, r_2, a_1, a_2) = (r^*, r^*, r^*, r^*)$, we compute the Jacobian matrix of the system (1) and (2) evaluated at that point:

$$J = \begin{pmatrix} -\frac{1}{\tau} & -\frac{\beta g}{\tau} & -\frac{\gamma g}{\tau} & 0 \\ -\frac{\beta g}{\tau} & -\frac{1}{\tau} & 0 & -\frac{\gamma g}{\tau} \\ \frac{1}{\tau_a} & 0 & -\frac{1}{\tau_a} & 0 \\ 0 & \frac{1}{\tau_a} & 0 & -\frac{1}{\tau_a} \end{pmatrix},$$

where:

$$g = \sigma'(I - (\gamma + \beta)r^*) \geq 0.$$

Because the system is invariant under the permutation $(r_1, r_2, a_1, a_2) \mapsto (r_2, r_1, a_2, a_1)$ (symmetry $1 \leftrightarrow 2$), the 4×4 Jacobian can be expressed, after a suitable change of coordinates, as a block-diagonal matrix.[1] In these coordinates, the Jacobian decouples into two 2×2 blocks:

$$A_{\text{sym}} = \begin{pmatrix} -\frac{1+\beta g}{\tau} & -\frac{\gamma g}{\tau} \\ \frac{1}{\tau_a} & -\frac{1}{\tau_a} \end{pmatrix}, \qquad A_{\text{antisym}} = \begin{pmatrix} -\frac{1-\beta g}{\tau} & -\frac{\gamma g}{\tau} \\ \frac{1}{\tau_a} & -\frac{1}{\tau_a} \end{pmatrix}.$$

Let us compute the characteristic polynomials for both cases. For the block A_{sym}:

$$\text{Tr}(A_{\text{sym}}) = -\frac{1+\beta g}{\tau} - \frac{1}{\tau_a}, \tag{9}$$

$$\det(A_{\text{sym}}) = \frac{1+\beta g+\gamma g}{\tau\tau_a}, \tag{10}$$

$$p_{\text{sym}}(\lambda) = \lambda^2 - \Big(-\frac{1+\beta g}{\tau} - \frac{1}{\tau_a}\Big)\lambda + \frac{1+\beta g+\gamma g}{\tau\tau_a}. \tag{11}$$

And for the block A_{antisym}:

$$\text{Tr}(A_{\text{antisym}}) = -\frac{1-\beta g}{\tau} - \frac{1}{\tau_a}, \tag{12}$$

$$\det(A_{\text{antisym}}) = \frac{1-\beta g+\gamma g}{\tau\tau_a}, \tag{13}$$

$$p_{\text{antisym}}(\lambda) = \lambda^2 - \Big(-\frac{1-\beta g}{\tau} - \frac{1}{\tau_a}\Big)\lambda + \frac{1-\beta g+\gamma g}{\tau\tau_a}. \tag{14}$$

The characteristic polynomial associated with the equilibrium point therefore factorizes as

$$P(\lambda) = p_{\text{sym}}(\lambda)\, p_{\text{antisym}}(\lambda).$$

Thus, the characteristic exponents associated with the equilibrium point (r^*, r^*, r^*, r^*) are:

$$\lambda^{\pm}_{\text{sym}} = \frac{1}{2}\left(-\frac{1+\beta g}{\tau} - \frac{1}{\tau_a} \pm \sqrt{\Big(-\frac{1+\beta g}{\tau} - \frac{1}{\tau_a}\Big)^2 - \frac{4(1+\beta g+\gamma g)}{\tau\tau_a}}\right),$$

$$\lambda^{\pm}_{\text{antisym}} = \frac{1}{2}\left(-\frac{1-\beta g}{\tau} - \frac{1}{\tau_a} \pm \sqrt{\Big(-\frac{1-\beta g}{\tau} - \frac{1}{\tau_a}\Big)^2 - \frac{4(1-\beta g+\gamma g)}{\tau\tau_a}}\right).$$

[1] Using the symmetric and antisymmetric perturbation variables $x_s = x_1 + x_2,\ x_a = x_1 - x_2,\ y_s = y_1 + y_2,\ y_a = y_1 - y_2$ for perturbations of the form $r_1 = r^* + x_1,\ r_2 = r^* + x_2,\ a_1 = r^* + y_1,\ a_2 = r^* + y_2$, the Jacobian J is transformed into a block-diagonal matrix whose diagonal blocks are the 2×2 matrices A_{sym} and A_{antisym}.

7.2 Stability Conditions

Let us study the stability of the equilibrium point.

First, we note that for the symmetric block, from expressions (9) and (10) the trace is always negative, and the determinant is always positive. Therefore, the corresponding eigenvalues have negative real parts, and the symmetric mode is always linearly stable.

On the other hand, for the antisymmetric block, we note that both the trace and the determinant can be positive, negative, or zero. The condition for the trace (12) to be positive, negative, or zero can alternatively be expressed in terms of the function

$$g_{trace}(I, \gamma, \beta) = \tau_a \beta g(I, \gamma, \beta) - (\tau_u + \tau), \tag{15}$$

whereas for the determinant (13) can be expressed in terms of

$$g_{det}(I, \gamma, \beta) = 1 + (\gamma - \beta) g(I, \gamma, \beta). \tag{16}$$

This implies that depending on the parameter values I, γ and β the linear stability associated with the antisymmetric eigenvalues will change. In general, we have the following possibilities:

- $\det(A_{\text{antisym}}) > 0$ (i.e., $g_{det} > 0$) and $\text{Tr}(A_{\text{antisym}}) < 0$ (i.e., $g_{trace} < 0$): The eigenvalues $\lambda^{\pm}_{\text{antisym}}$ have negative real part and the antisymmetric mode is linearly stable. This is the only case in which (r^*, r^*, r^*, r^*) is asymptotically stable for the nonlinear system.

- $\det(A_{\text{antisym}}) > 0$ (i.e., $g_{det} > 0$) and $\text{Tr}(A_{\text{antisym}}) > 0$ (i.e., $g_{trace} > 0$): The eigenvalues $\lambda^{\pm}_{\text{antisym}}$ have positive real part and the antisymmetric mode is linearly unstable. In this case, (r^*, r^*, r^*, r^*) is a saddle point (hence is unstable) with 2-dimensional stable and unstable invariant manifolds.

- $\det(A_{\text{antisym}}) < 0$ (i.e., $g_{det} < 0$): The eigenvalues $\lambda^{\pm}_{\text{antisym}}$ are real, with $\lambda^{+}_{\text{antisym}}$ positive and $\lambda^{-}_{\text{antisym}}$ negative. In this case, (r^*, r^*, r^*, r^*) is a saddle point (hence is unstable) with a 3-dimensional stable invariant manifold and a 1-dimensional unstable invariant manifold.

In addition, the following bifurcations are possible:

- $\det(A_{\text{antisym}}) > 0$ (i.e., $g_{det} > 0$) and $\text{Tr}(A_{\text{antisym}}) = 0$ (i.e., $g_{trace} = 0$): The eigenvalues $\lambda^{\pm}_{\text{antisym}}$ have zero real part and nonzero imaginary part. Hence, the antisymmetric linearized system undergoes an Andronov-Hopf bifurcation.

- $\det(A_{\text{antisym}}) = 0$ (i.e., $g_{det} = 0$): In this case, the determinant of the Jacobian vanishes, and since the system has the symmetry $1 \leftrightarrow 2$, it undergoes a Pitchfork bifurcation.

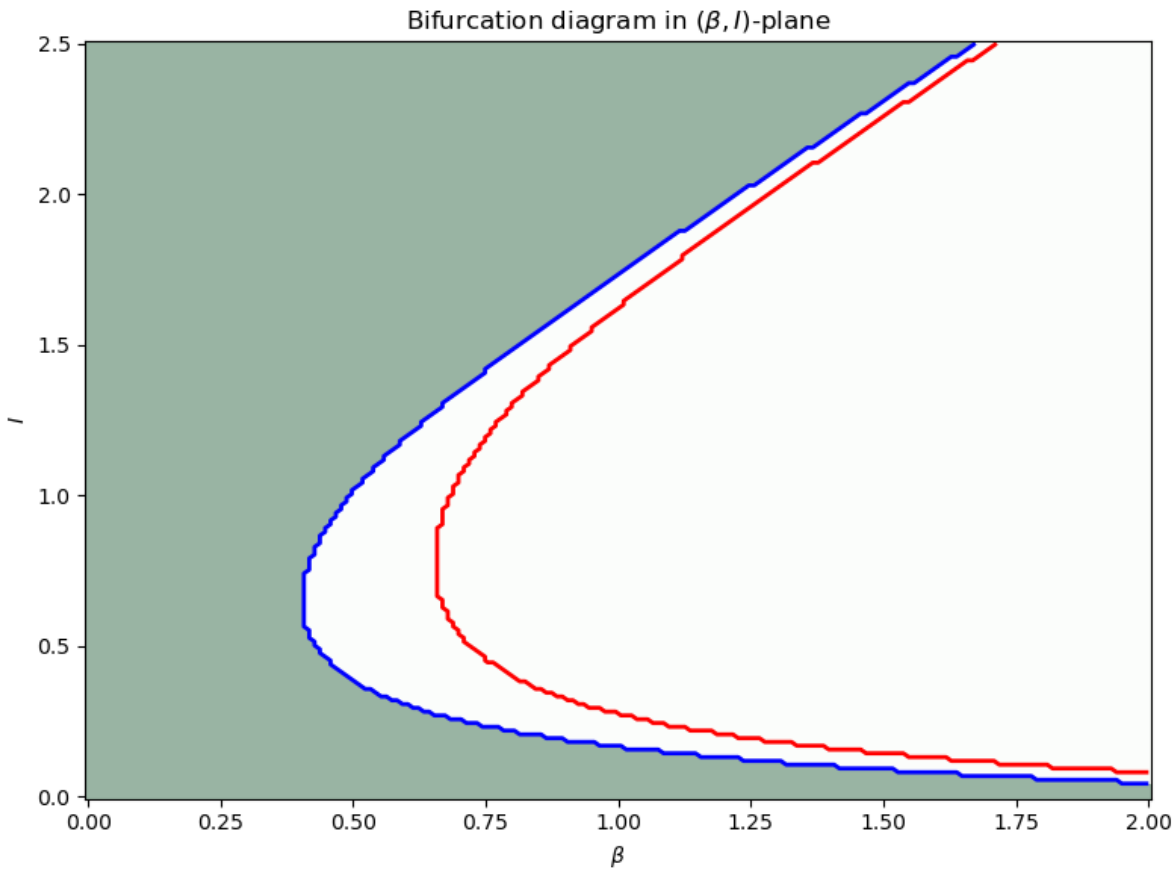


Figure 8: Bifurcation diagram in the (β, I)-plane, for a fixed $\gamma = 0.5$. The green region indicates the parameter region in which all eigenvalues have negative real parts, corresponding to an asymptotically stable equilibrium. The white region indicates where one or more eigenvalues have positive real parts, corresponding to an unstable equilibrium. The blue and red lines represent Hopf and Pitchfork bifurcations, respectively.

8 Stochastic System

In this final section, we note that neuronal populations are inherently noisy, and therefore a stochastic system of equations may be used to describe binocular rivalry.

We introduce randomness into the set of differential equations describing the firing rates of populations 1 and 2 through uncorrelated colored noises η_1 and η_2:

$$\begin{cases} \tau \dfrac{dr_1}{dt} = -r_1 + \sigma(-\gamma a_1 - \beta r_2 + I_1 + \eta_1), \\ \tau \dfrac{dr_2}{dt} = -r_2 + \sigma(-\gamma a_2 - \beta r_1 + I_2 + \eta_2), \end{cases} \tag{17}$$

$$\begin{cases} \tau_a \dfrac{da_1}{dt} = -a_1 + r_1, \\ \tau_a \dfrac{da_2}{dt} = -a_2 + r_2, \end{cases} \tag{18}$$

where the noises are solutions of the stochastic differential equations:

$$\begin{cases} \dfrac{d\eta_1}{dt} = -\dfrac{\eta_1}{\tau_n} + \xi(t), \\ \dfrac{d\eta_2}{dt} = -\dfrac{\eta_2}{\tau_n} + \xi(t), \end{cases} \tag{19}$$

and $\xi(t)$ is the stochastic term, which is modelled as a white noise of amplitude $\rho = 0.05$ and timescale of $\tau_n = 10$ms.

One of our focuses in this section is to simulate the stochastic evolutions of $r_1(t)$ and $r_2(t)$. The most effective approach is to proceed numerically by discretizing time with a step size $\Delta t = t_{i+1} - t_i$. We will simulate the system for a time extension of $t_f = 10$s, which corresponds to $N = 100,000$ time steps of size $\Delta t = 0.1$ ms.

In this case, the system of differential equations becomes a system of recurrence relations that can be solved numerically for a given set of initial conditions. Specifically, we have:

$$\begin{cases} r_1(t_{i+1}) = r_1(t_i) + \big[- r_1(t_i) + \sigma(-\gamma a_1(t_i) - \beta r_2(t_i) + I_1 + \eta_1(t_i))\big] \dfrac{\Delta t}{\tau}, \\ r_2(t_{i+1}) = r_2(t_i) + \big[- r_2(t_i) + \sigma(-\gamma a_2(t_i) - \beta r_1(t_i) + I_2 + \eta_2(t_i))\big] \dfrac{\Delta t}{\tau}, \end{cases} \tag{20}$$

$$\begin{cases} a_1(t_{i+1}) = a_1(t_i) + \big[- a_1(t_i) + r_1(t_i)\big] \dfrac{\Delta t}{\tau_a}, \\ a_2(t_{i+1}) = a_2(t_i) + \big[- a_2(t_i) + r_2(t_i)\big] \dfrac{\Delta t}{\tau_a}, \end{cases} \tag{21}$$

$$\begin{cases} \eta_1(t_{i+1}) = \eta_1(t_i) - \eta_1(t_i) \dfrac{\Delta t}{\tau_n} + \rho\, z(t_i)\sqrt{\Delta t}, \\ \eta_2(t_{i+1}) = \eta_2(t_i) - \eta_2(t_i) \dfrac{\Delta t}{\tau_n} + \rho\, z(t_i)\sqrt{\Delta t}. \end{cases} \tag{22}$$

The last pair of equations corresponds to the Euler–Maruyama method. Randomness is introduced into the system at each time step through the random variable $z(t_i) \in \mathcal{N}(0,1)$, which follows a standard normal distribution. The terms multiplying the random variable $z(t_i)$ are the amplitude ρ and the time scaling $\sqrt{\Delta t}$, which arise from the theory of stochastic differential equations.

8.1 Stochastic Simulations

We can simulate the stochastic processes $r_1(t)$ and $r_2(t)$ for certain stimuli $I_1 = I_2 = I$ according to the previous expressions. For instance, for $I = 0.8$ (as in Figure 1), we obtain stochastic time evolutions like the one shown in Figure 9.

If we compare this stochastic time evolution with the one obtained in the deterministic case (Figure 1), we observe important differences. On the one hand, the dynamics in the deterministic case were oscillatory, with equal periodic dominance times for the firing rates of populations 1 and 2. On the other hand, in the stochastic system, both firing rates also exhibit periods of dominance (oscillatory dynamical regime), but these dominance durations are variable and take random values.

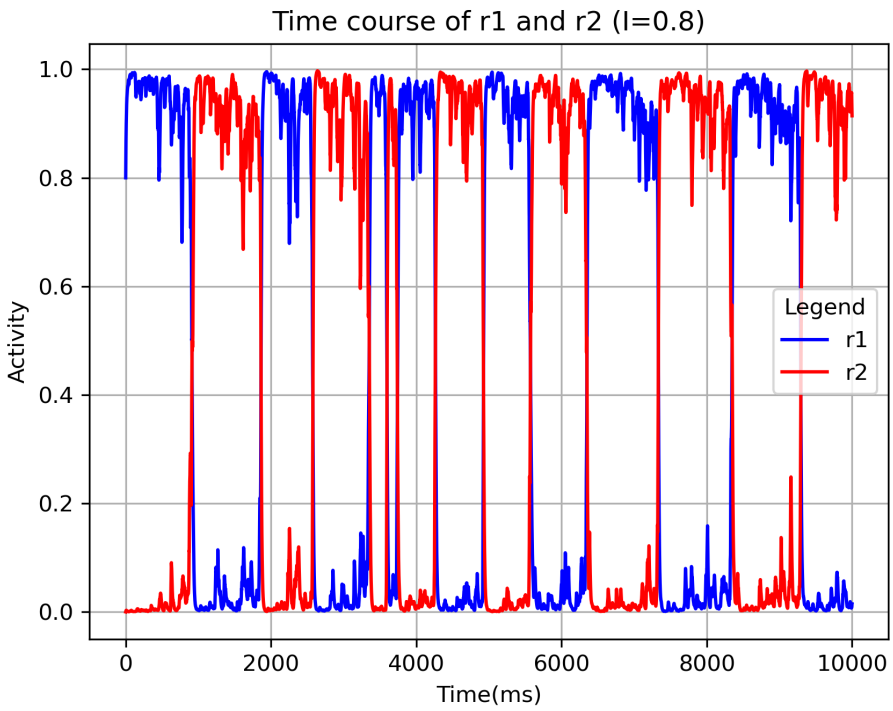


Figure 9: Stochastic simulation of the time evolution of the firing rates of the populations in response to a stimulus $I = 0.8$.

8.2 Dominance Duration Distribution

Our goal in this subsection is to analyse the probability distribution followed by the previously introduced random dominance durations.

In order to quantify the dominance durations, we introduce the dominance variable

$$d(t) = r_1(t) - r_2(t)$$

and the rule that population 1 dominates at time t if $d(t) > \omega$, whereas for $d(t) < -\omega$ population 2 dominates. We take a dominance threshold $\omega = 0.3$.

We also introduce two random variables T_1 and T_2, which correspond to the dominance durations of populations 1 and 2 respectively.

By simulating multiple stochastic time evolutions of the system and applying the dominance rule at each time step t_i, we obtain statistical samples of the dominance durations, namely

$$(T_1^1, T_1^2, \ldots, T_1^n) \quad \text{and} \quad (T_2^1, T_2^2, \ldots, T_2^m).$$

We may fit the dominance durations samples by a gamma distribution Gamma(α, λ), with pdf:

$$f(x; \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0 \tag{23}$$

The distribution parameters for each sample of T_1 and T_2, namely α (shape parameter) and λ (scale parameter), can then be obtained using their corresponding maximum likelihood estimators. A precise derivation of these estimators for the gamma distribution is provided in the Appendix.

Finally, we include some histograms of the dominance durations for different stimuli values I, along with the gamma distribution fitted to the data using the procedure described above.

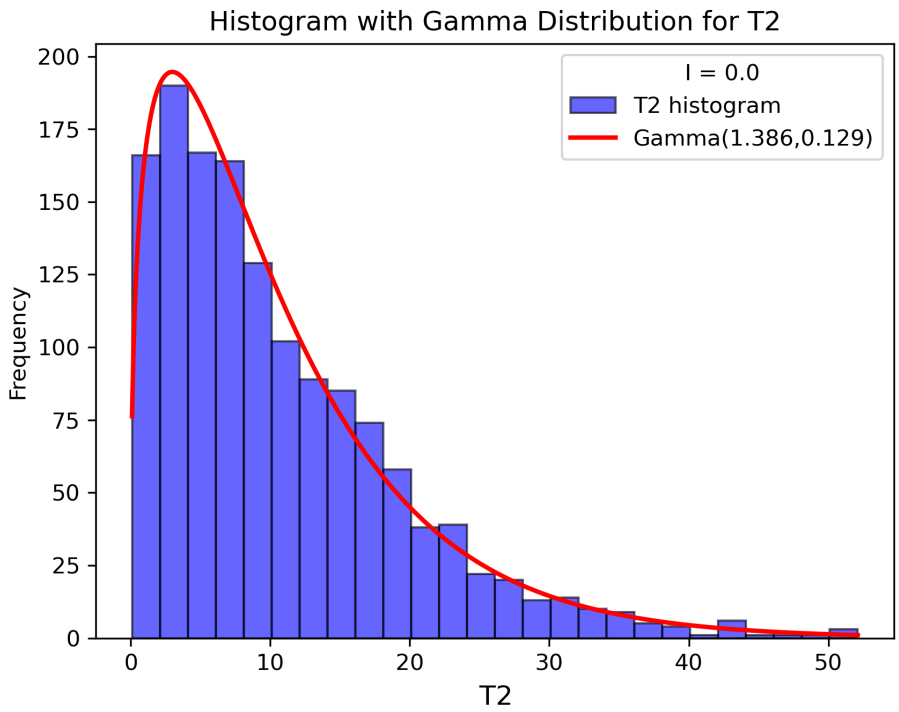


Figure 10: Histogram of the dominance duration T_2 (in ms), and gamma distribution fitted using point estimates from the histogram data, for a non-stimulated system.

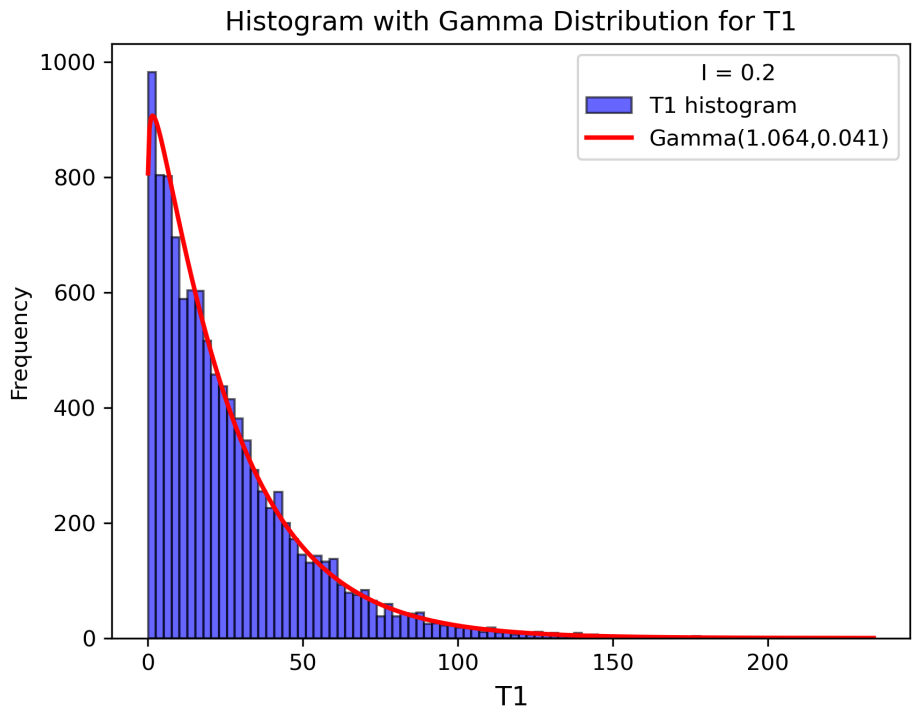


Figure 11: Histogram of the dominance duration T_1 (in ms), and gamma distribution fitted using point estimates from the histogram data, for a stimulated system with $I = 0.2$.

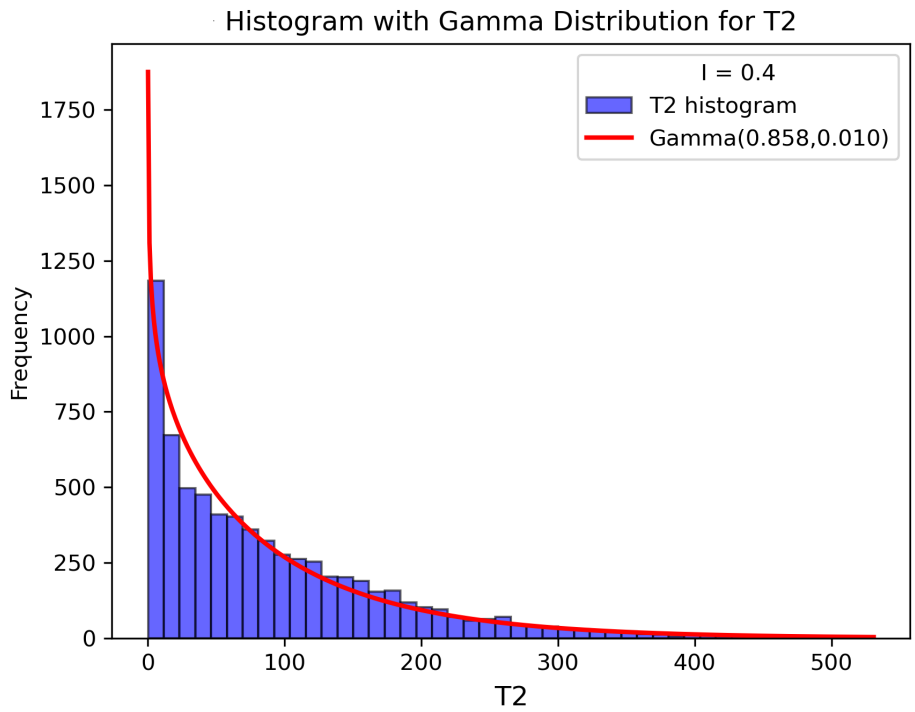


Figure 12: Histogram of the dominance duration T_2 (in ms), and gamma distribution fitted using point estimates from the histogram data, for a stimulated system with $I = 0.4$.

9 Conclusions

In this work, we examined both the deterministic binocular rivalry model and its stochastic counterpart. For the deterministic system, we explored the oscillatory behaviour arising from neural competition and showed how these oscillations can disappear as the model transitions toward a winner-take-all regime. Particular attention was devoted to the role of the adaptation timescale τ_a, which proved essential for generating slow adaptation dynamics and determining the existence and stability of equilibrium states.

For the stochastic formulation, we simulated the temporal evolution of the rating variables and analysed the resulting probability distribution of dominance durations. These simulations allowed us to assess how intrinsic noise shapes perceptual switching and how well the model reproduces known features of binocular rivalry, such as variability in dominance periods.

Appendix

Maximum Likelihood Estimator for a Gamma Distribution

Let $X_1, \ldots, X_n$ be a simple random sample following a gamma distribution Gamma(α, λ), with the pdf presented in expression (23). For each observation $\mathbf{x} = (x_1, \ldots, x_n) \in [0, \infty)^n$, the likelihood function is expressed as:

$$L(\alpha, \lambda \mid \mathbf{x}) = \frac{\lambda^{n\alpha}}{\Gamma(\alpha)^n} (\prod_{i=1}^{n} x_i)^{\alpha-1} \exp\left(-\lambda \sum_{i=1}^{n} x_i\right)$$

and its logarithm as:

$$\ln L(\alpha, \lambda \mid \mathbf{x}) = n\alpha \ln(\lambda) - n \ln(\Gamma(\alpha)) + (\alpha - 1) \sum_{i=1}^{n} \ln(x_i) - \lambda \sum_{i=1}^{n} x_i.$$

We can obtain the maximum likelihood estimators (MLE) of α and λ, $\hat{\alpha} = \hat{\alpha}(\mathbf{x})$ and $\hat{\lambda} = \hat{\lambda}(\mathbf{x})$, by using iterative optimization:

$$L(\alpha, \lambda \mid \mathbf{x}) \leq L(\alpha, \hat{\lambda} \mid \mathbf{x}) \leq L(\hat{\alpha}, \hat{\lambda} \mid \mathbf{x}).$$

Let $\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ and $\overline{\ln x} = \frac{1}{n} \sum_{i=1}^{n} \ln(x_i)$. Then, by finding the maximum as $\frac{\partial}{\partial \lambda} \ln L(\alpha, \lambda \mid \mathbf{x}) = 0$, we obtain the relation:

$$\hat{\lambda} = \frac{\alpha}{\overline{x}}.$$

Now, if we impose $\frac{\partial}{\partial \alpha} \ln L(\alpha, \hat{\lambda} \mid \mathbf{x}) = 0$, we obtain that $\hat{\alpha}$ must be a zero of

$$h(\alpha) := \ln(\alpha) - \psi(\alpha) + \overline{\ln x} - \ln \overline{x},$$

where $\psi(\alpha)$ is the Digamma function. Therefore, the determination of the MLEs $\hat{\lambda}$ and $\hat{\alpha}$ reduces to finding an appropriate zero of $h(\alpha)$.

We may use Newton's method to find a zero of $h(\alpha)$ numerically, using the recurrence

$$\alpha_{n+1} = \alpha_n - \frac{h(\alpha_n)}{h'(\alpha_n)},$$

where $h'(\alpha) = \frac{1}{\alpha} - \psi'(\alpha)$, which converges to a root of h for an appropriate initial guess.

To ensure the convergence of Newton's method, we take as initial guess the estimator of α given by the moment's method:

Since the mean μ and variance σ^2 of the gamma distribution are given by relations

$$\mu = \frac{\alpha}{\lambda}, \qquad \sigma^2 = \frac{\alpha}{\lambda^2}.$$

then we have

$$\alpha = \frac{\mu^2}{\sigma^2}.$$

By making use of the Law of Large Numbers, we obtain an initial guess

$$\alpha_0 = \frac{\overline{x}^2}{\mathrm{var}(\mathbf{x})}.$$

where $\mathrm{var}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$.